Ashoka University | Department of Economics

# ECO-3554/ECO-4554 Algorithmic Economics

Monsoon Semester 2026 (Aug–Dec)

| | |
|---|---|
| **Instructor:** | Sumit Goel |
| **Email:** | sumit.goel@ashoka.edu.in |
| **Office Hours:** | TBD |
| **Teaching Fellow:** | Akash Sen |
| **TF Email:** | imsenakash@gmail.com |
| **TF Office Hours:** | TBD |
| **Lectures:** | Tu-Th 10:10 AM –11:40 AM in AC-04-LR-303 |
| **Course Webpage:** | goelsumit.com/algoecon.html |

## Course Description

The course is an introduction to economic design, with applications to resource allocation, fair division, social choice, and auctions. In these contexts, we will formulate design goals, discuss rules that achieve them, and study what can be done when these goals are incompatible. We will consider social objectives such as efficiency and fairness, as well as private objectives such as revenue maximization, subject to incentive, computational, and communication constraints.

## Learning Objectives

- Become familiar with the tools of microeconomics that guide modern economic design.
- Identify incentive issues, and develop a working knowledge of mechanism design.
- Recognize the computational and communicational issues in economic environments.

## Prerequisites

There are no formal course prerequisites. The content of the course is, however, formal and mathematical. Students should be comfortable with formal mathematical proofs and will be expected to write proofs on their own.

## Evaluation

This course is lecture based, and student assessment focuses on problem solving. The final course grade will be composed of the following components:

| Component | Weight |
|---|---|
| Assignments/Quizzes | 40% |
| Midterm exam | 30% |
| Final exam | 30% |

The numeric score will be converted to a letter grade according to the following scheme:

| A | A− | B+ | B | B− | C+ | C | C− | D+ | D | F |
|---|---|---|---|---|---|---|---|---|---|---|
| >95 | 90–95 | 85–90 | 80–85 | 75–80 | 70–75 | 65–70 | 60–65 | 55–60 | 50–55 | <50 |

## Attendance

Regular attendance in this course is expected and highly recommended.

## Course Outline

| **Weeks** | **Topics** |
|---|---|
| Week 1 | **Aggregating preferences:** efficiency, incentives, impossibility |
| Weeks 2–3 | **Escaping impossibilities:** domain restrictions, complexity barriers |
| Weeks 4–5 | **Quasilinear domain:** VCG, approximation ratio |
| Weeks 6–7 | **Auctions:** revenue maximization, combinatorial auctions |
| Weeks 8–9 | **Fair division (divisible goods):** cake cutting, rent division |
| Weeks 10–11 | **Fair division (indivisible goods):** Nash welfare, randomization |
| Weeks 12–13 | **Cooperative games:** Shapley value, feature attribution in ML |

## Readings

You may find the following sources useful, which I use to prepare the lectures:

- Lecture notes by Fedor Sandomirskiy
- Lecture notes by Ariel Procaccia

I also borrow ideas from Federico Echenique's course on Algorithmic Economics.

*This syllabus is subject to change at the discretion of the instructor.*